Military Institute of Science and Technology

FINAL PROJECT REPORT

Unmanned Ground Reconnaissance Vehicle

A four-wheel Wi-Fi controlled vehicle for remote inspection, live video and side-distance logging.

PROJECT INFORMATION		
Department	Computer Science and Engineering	
Course	Microprocessor, Microcontroller and Assembly Language	
Batch	CSE-24, level-3, Section-A	
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1 Project Title

Unmanned Ground Reconnaissance Vehicle (UGRV)

2 Project Description

Purpose and idea

We built this small ground vehicle to let a person inspect a nearby space without standing beside the object or area being observed. A phone or laptop becomes the control station: we can drive the car, watch a live camera feed, adjust the camera angle, and check distance readings through a local web page. The idea is deliberately practical. A compact, low-cost platform can travel under furniture, approach a hard-to-reach corner, or enter a place where we would prefer to observe first and move closer later.

Our model uses a four-wheel acrylic chassis with four geared DC motors. We borrowed the main ESP32 development board from our laboratory for this build. It handles the vehicle controls, web dashboard, distance sensing, and on-board logging. A separate ESP32 camera module supplies the video stream over the same local Wi-Fi network. Two rechargeable 3.7 V cells form the nominal 7.4 V source for the drive system, while a buck converter provides a regulated 5 V rail for the electronics. This split lets us operate the motors and low-voltage control devices from one battery pack while giving the camera the stable supply it needs.

Scope of the present model

We designed the vehicle for short-range, local reconnaissance and demonstration. It brings remote movement, live video, side-distance sensing, and a record of measurements together on one platform. Four ultrasonic modules are positioned in two pairs, with two on the left and two on the right. The main controller processes left and right distance channels and shows their readings on the dashboard. The operator uses those readings alongside the camera view to judge nearby space and guide the car.

The main ESP32 creates a local access point, so a nearby operator can use the car without a separate router or internet connection. Its dashboard provides movement commands, a camera view, recent range data, and controls for the servo. We use the term reconnaissance because the operator sees and measures the surroundings while retaining direct control of the vehicle. This makes the model relevant to inspection, classroom demonstrations, and early experiments in mobile robotics.

We chose a small wheeled car because it is easy to carry, repair, and test indoors. The operator can stop at a doorway, look around through the camera, compare the left and right distances, and decide whether there is enough room to continue. That sequence is the central idea of the project: the vehicle extends our view into a nearby space while a person still decides where it should go. The model is a teaching platform as well, since each function can be examined separately: drive control, sensing, wireless video, power conversion, and data recording.

3 Component List and Cost

Table 1 presents the components and a project cost estimate in Bangladeshi taka. We borrowed the main ESP32 from our laboratory, so its replacement value is included in the full system cost but excluded from the cost of parts supplied outside the lab. The prices provide a consistent budget for assembling the same type of vehicle.

Component and model	Qty	Unit Tk	Total Tk
4WD acrylic chassis kit with 4 BO gear motors and wheels	1 kit	999	999
ESP32 DevKit type main controller; borrowed from lab	1	490	490
ESP32-CAM with GC2145 camera	1	1,370	1,370
L298N dual H-bridge motor driver	1	480	480
HC-SR04 ultrasonic sensor module	4	120	480
SG90 micro servo	1	140	140
LM2596 style adjustable buck converter	1	85	85
18650 Li-ion cell, 3.7 V nominal	2	260	520
Two-cell 18650 holder	1	50	50
Rocker power switch	1	18	18
DC barrel jack	1	25	25
ESP32 mounting/breakout board	1	260	260
Jumper wires	1 set	140	140

Table 1 Indicative component cost for the current model

Estimated full system value: Tk 5,137. After excluding the lab-supplied main ESP32, the value of the remaining parts is Tk 4,647.

Other Mechanisms

The L298N is a dual H-bridge. We use one channel for the two motors on the left and one for the two motors on the right. By changing the polarity and duty cycle of each channel, the main ESP32 sets direction and speed. The LM2596 style buck converter reduces the battery voltage to a regulated 5 V output for the electronics, separating the motor supply from the control supply.

The ESP32-CAM combines a camera sensor and Wi-Fi controller on a separate board. It sends video to the browser through the local network, so no video data wire is needed between the two ESP32 boards. The HC-SR04 measures distance by timing an ultrasonic echo. Each complete module has a VCC, GND, TRIG, and ECHO connection and contains two round transducer heads. LittleFS is flash storage inside the main ESP32; we use it to retain a CSV log of distance readings without adding an external memory card.

The four wheel motors are grouped by side because the L298N provides two drive channels. This arrangement keeps the control circuit simple and gives us a clear way to steer by varying left and right motor motion. We mounted the servo near the camera so that changing its angle changes what the operator can inspect without rotating the entire car. The acrylic chassis makes the wiring and parts visible during demonstration.

4 Working Methodology

We begin by switching on the battery supply. The motor driver receives the nominal 7.4 V drive rail and the converter produces 5 V for the control boards and accessories. Once the main ESP32 starts its access point, the operator joins the UGRV CAR Wi-Fi network and opens the dashboard at 192.168.4.1. Browser commands reach the main ESP32, which changes the L298N inputs to move the car. The left and right motor pairs can then turn in the same direction for forward or reverse travel, or in different directions for steering.

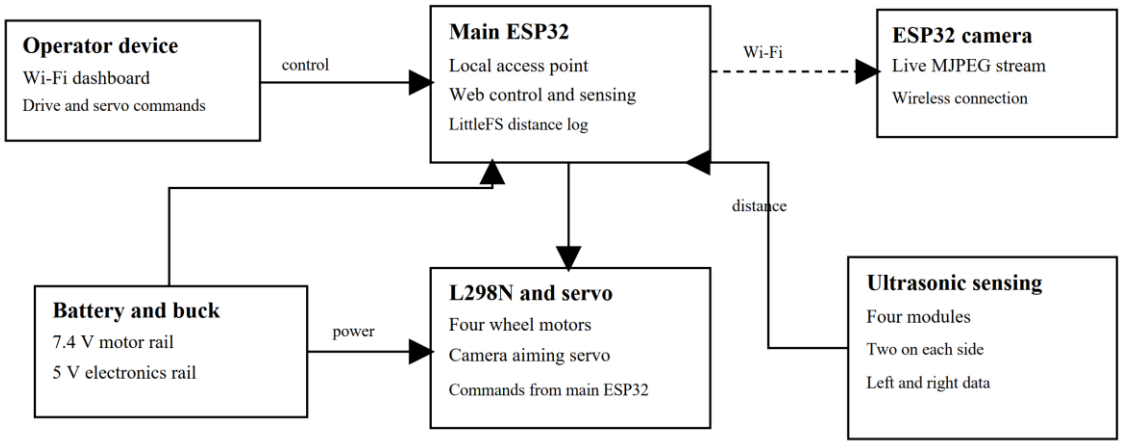


Figure 1 Control, power and data paths in the current model

Sensing, logging and video

The four ultrasonic modules face outward in two pairs, with two on each side of the car. The controller alternates measurements between its left and right channels, filters invalid echoes, and displays the latest distances on the dashboard. After the browser supplies time information, the code writes a paired reading to a CSV file about every 20 seconds. The dashboard shows recent entries and provides a download route. This creates a useful record of the route surroundings for later review.

The camera board connects wirelessly to the main ESP32 access point, registers its network address, and serves an MJPEG stream. The browser loads that stream directly from the camera's local address. The SG90 servo responds to the main ESP32 and lets us adjust the viewing angle. Because the camera is a separate device, we can inspect the video path and the vehicle-control path independently when troubleshooting.

We keep the operator in control of motion. The live picture shows what lies ahead, while sonar readings provide a distance cue for nearby side objects. The firmware includes a short motor-command timeout that stops the drive when browser control messages end. Together, the camera, range data, and direct steering controls give the operator a clear view of the vehicle's immediate surroundings.

Power distribution

Two 3.7 V cells are connected in series to provide about 7.4 V nominal for the drive system. The motor driver takes the battery supply, while the buck converter produces the 5 V rail used by the controller, camera board, servo, and sensors. The switch controls the battery line. All modules share a common ground so that control signals have the same electrical reference. The DC barrel connector provides the external power connection shown on the vehicle.

5 Full Circuit Schematic

Figure 2 shows the system power path, motor driver, main ESP32, camera module, servo, and the four ultrasonic modules. The GPIO labels follow the main controller program. Two modules are positioned on each side of the car, while the software uses left and right distance channels for dashboard display and logging.

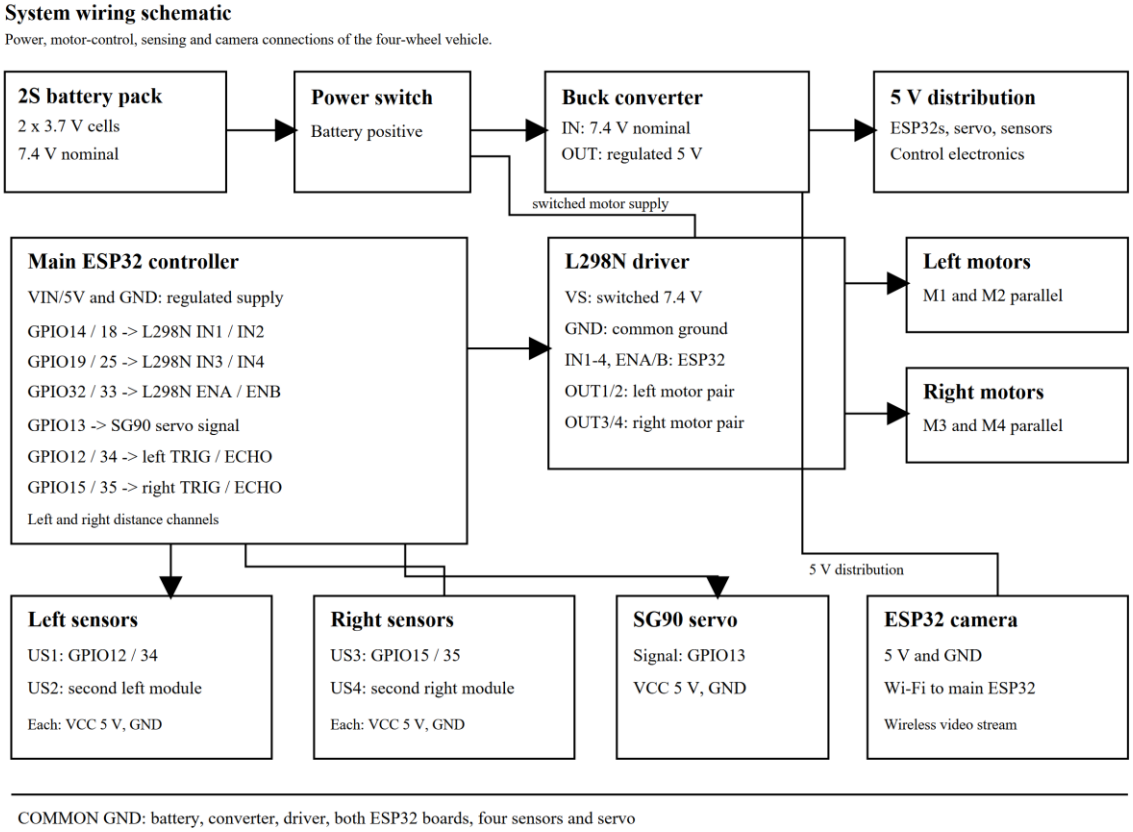


Figure 2 Power and signal connections of the current vehicle

Main ESP32 GPIO	Device connection	Function
14, 18	L298N IN1, IN2	Left motor pair direction
19, 25	L298N IN3, IN4	Right motor pair direction
32, 33	L298N ENA, ENB	PWM speed control
13	SG90 signal	Camera-angle servo
12, 34	Left channel TRIG, ECHO	Left distance measurement
15, 35	Right channel TRIG, ECHO	Right distance measurement

Table 2 Signal connections identified in the supplied main ESP32 firmware

The L298N OUT1/OUT2 terminals feed the two left motors in parallel, while OUT3/OUT4 feed the two right motors in parallel. GPIO14, 18, 19, and 25 set motor direction, and GPIO32 and 33 supply the two speed-control signals. GPIO13 controls the camera servo. The battery supplies the L298N motor input and buck converter input; the converter provides the 5 V electronics rail. The camera joins the main ESP32 over Wi-Fi and sends its stream directly to the browser.

The left and right sensing channels use GPIO12/34 and GPIO15/35 for trigger and echo signals. The controller processes these measurements, displays the latest distances, and stores timed readings in LittleFS. This arrangement ties vehicle motion, observation, and data capture together in one operator dashboard.

6 Future Implications and Improvements

Independent four-sensor readout

We can extend the present left-and-right display into four independently labelled readings: left front, left rear, right front, and right rear. The controller would trigger each module in sequence to prevent one sensor from hearing another sensor's pulse. Showing all four values on the dashboard would make it easier to tell whether an obstacle is beside the front or rear wheel. We would calibrate the readings against measured distances before using them for automatic decisions.

Assisted obstacle avoidance

At present, we make driving decisions ourselves by watching the stream and reading distances. A practical next step is an assisted stop: if a measured distance falls below a chosen threshold, the controller would reduce speed or stop that side of the drive. We would test the threshold at several speeds and surfaces and keep a clear manual override. Once that behavior is reliable, we could explore automatic turning around an obstacle. This gradual approach would make the car safer to operate without assuming that a single noisy echo is enough to plan a route.

Mapping and position awareness

A distance log by itself does not tell us where each reading was taken. To build a usable map, we would add wheel encoders and an inertial measurement unit so that we can estimate distance travelled and heading. We could then combine time-stamped position estimates with the four range readings and plot obstacles relative to the vehicle path. Such a map would still drift over time, so repeated trials and reference points would be needed before claiming reliable navigation.

Electrical reliability and longer operation

For longer operating sessions, we would add battery-voltage monitoring and measure current during normal travel and motor start-up. We could then compare the power drawn by the four motors, camera, and servo with the chosen battery capacity. A more efficient motor driver would reduce heat and wasted energy. A dedicated charging and protection assembly, fuse, and clearly marked power connector would make the electrical system easier to maintain.

Video, enclosure and access

For use outside the lab, we would secure the exposed wiring, protect the camera and sensors from impacts, and check Wi-Fi range where the car is actually operated. We could improve low-light observation with suitable lighting and camera tuning. The current local dashboard is useful for a prototype; before wider use, we would add authentication, make connection loss visible to the operator, and retain the motor timeout that stops unattended motion. These changes would make the vehicle more dependable while keeping the basic purpose of remote inspection.

In summary, we built a platform for supervised local reconnaissance with four driven wheels, live video, side-distance sensing, and ESP32 logging. The same architecture gives us a practical starting point for independent four-sensor display, assisted obstacle avoidance, and mapping.

7 Conclusion

We built the Unmanned Ground Reconnaissance Vehicle to inspect nearby spaces while operating from a safe distance. Our four-wheel vehicle combines remote driving, live camera video, side-distance sensing, camera-angle control, and data logging in one web dashboard. The main ESP32 coordinates the controls and measurements, while a separate ESP32 camera provides the video feed over the vehicle's local Wi-Fi network.

The project gave us a practical way to bring together mechanical design, power distribution, sensors, and wireless communication. It also provides a foundation for our next steps: displaying readings from all four ultrasonic sensors independently, adding assisted obstacle avoidance, and developing position-aware mapping.